

Oleksii (Alex) Lavrenin — Project Summaries

Models, training, evaluation, and results across eight projects

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Every figure below is a measured number from the project's own artifacts — metrics JSONs, papers, or diagnostic outputs. Where a project has **no trained model or no formal evaluation**, this document says so rather than borrowing credibility from the ones that do.

uSport.ai

KNOWLEDGE GRAPH · AGENTS · RAG

2023 – present

AI recruiting platform matching student-athletes to NCAA coaches. Ten product surfaces over one substrate: a Neo4j graph of 8M+ athletes, 51K coaches, 176K schools, 15M+ swim results and 1,400+ NCAA rules, with seven vector indexes native to the same database.

MODELS	Claude (all agents), GPT-4o (re-ranking, scoring, email), text-embedding-3-small (1536-d). Task models documented separately below.
TRAINING	<i>No model trained at platform level</i> — retrieval and agent behavior are grounded by Cypher traversals and vector search over indexed content.
EVALUATION	Per-subsystem (see Two-Tower Retrieval and Career Projection). Agent quality measured through Agent Gym's judge rubric.
RESULTS	In production. 27.9M-node graph, 8.4M athlete profiles, 7 vector indexes, 15+ agent tools across 6 RAG chains.

Agent Gym

MULTI-AGENT SIMULATION

2026

Simulation environment where a coach agent and an athlete agent negotiate across three structured phases against real profiles from the graph. A separate model scores every turn; high-scoring episodes become retrieval context for the production assistant.

MODELS	Agents: Claude Sonnet 4.5 . Judge: GPT-5 (forced for auto-training) or Claude. Embeddings: text-embedding-3-small.
TRAINING	No gradient updates. Improvement is in-context: episodes scoring ≥ 0.6 total reward and ending in a successful terminal state are retrieved as few-shot examples for later episodes and for the live assistant. Cloud Scheduler generates 1 episode/hour unattended.
EVALUATION	Cross-model LLM judge, phase-aware, four weighted dimensions per turn — offer-probability delta 50%, factual grounding 20%, regulatory compliance 20%, sentiment 10%. Solo-research mode uses a separate 35/25/20/20 rubric.
RESULTS	3 episode modes in production; episodes stored as graph nodes with 1536-d embeddings and consumed by the shipped recruiting assistant.

Bidirectional coach–athlete matching for college swimming recruiting. UC Berkeley DATASCI 266 (NLP with Deep Learning). Tests whether decoder-LLM normalization of inputs beats contrastive fine-tuning.

MODEL	BAAI/bge-large-en-v1.5 (1024-d, 512 tokens), shared by both towers. Plus T5-small for match explanations.
TRAINING	Contrastive fine-tuning, MultipleNegativesRankingLoss , 3 epochs, 1,196 triplets. Hard negatives = coaches at schools swimming the same events where the athlete is <i>not</i> competitive. Ground truth from 22,974 benchmark-derived pairs.
EVALUATION	70/30 split by swimmer (511 train / 220 test, disjoint — no leakage). P@K, MRR, nDCG@10, plus negative-pair AUC-ROC and bootstrap CIs.
RESULTS	Forward MRR 0.304 → 0.883 from fine-tuning (2.9×); LLM-normalization treatment 0.909 but with overlapping CIs — fine-tuning is the effect, normalization is not . Top-1 hit rate 83.6%. Reverse direction 0.045 → 0.202 (4.5× and still weak — coach profiles carry little distinctive text). Negative separation 0.015 → 0.086, AUC 0.633 → 0.666. T5 explanations: ROUGE ≈ 0.53 on n=2 — illustrative only.

Career Projection — NBA APPLIED ML · SPORTS ANALYTICS

2026

Three separate models answering three different questions about a basketball career. The pipeline covers eight sports; this section uses basketball only, since it is the deepest and every figure below is quoted from a committed metrics JSON.

DATA	5,368 NBA players indexed from Basketball-Reference season files — 4,452 retired with complete career data, 916 still active. Median career 4.0 years, mean 5.29, p90 12.0; 41.3% last two years or fewer , which is the attrition the models are trying to anticipate. High-school box scores come from a MaxPreps scrape; recruiting overlays from 247Sports, On3, ESPN-100 and McDonald's All-American.
MODEL 1 HS → NBA	Question: will this high-school player ever reach the NBA? Model: logistic regression and gradient boosting, compared. 22 features — 13 MaxPreps box-score fields (PPG, RPG, APG, SPG, FG%, 3P%, FT%, TS%, eFG%, height, weight, games, seasons) plus 9 recruiting-overlay fields. Data: 705 players — 364 who reached the NBA, 341 D1 college players who did not. Result: gradient boosting CV ROC-AUC 0.795 ± 0.032, logistic regression 0.766 ± 0.025. Gradient boosting: recall 0.670, precision 0.772.
MODEL 2 COLLEGE → NBA	Question: given a college career, will this player play in the NBA? Model: logistic regression (selected) vs gradient boosting. 7 features — college per-game points, rebounds, assists, FG%, 3P%, FT%, minutes. Data: 361 players — 166 NBA, 195 non-NBA. Result: ROC-AUC 0.910, recall 0.905, precision 0.776. <i>This number is inflated and the pipeline says so</i> — see Honest limits below.
MODEL 3 NBA LONGEVITY	Question: how many years will this NBA career last? Model: GradientBoostingRegressor (regression, not classification). Features: rookie year, position bucket, era. Data: 4,452 retired players — 3,561 train / 891 test. Result: MAE 3.63 years against a 3.85-year mean-predictor baseline; R² 0.077.
OUTPUT	Models 1 and 2 output a calibrated probability per player plus a ranked list. Model 3 outputs a per-player record — predicted total career years, years elapsed, projected years remaining, and a projected end year. It is scored over 120 active players ; a typical row reads <code>predicted_career_years 6.69 · years_in_so_far 1 · projected_remaining 5.69 · projected_end_year 2032</code> . Across all 120 the prediction takes only four distinct values , set entirely by position bucket — wings 6.69, bigs 6.49, guards 4.90, forwards 4.24 — with only the end year moving inside a bucket. That is what R ² = 0.077 looks like as output rather than as a statistic, and it is why the product shows a range rather than a point estimate. Logistic-regression coefficients are exported to JSON so the product scores new inputs in the browser with no model server.

HOW
ACCURACY IS
ESTABLISHED

- 1. Held-out data, never training data.** 5-fold stratified cross-validation; every AUC above is the mean across folds with its standard deviation, so run-to-run variance is visible rather than hidden by one lucky split.
- 2. Against a baseline, not in isolation.** The longevity model is only meaningful next to the mean predictor — 3.63 vs 3.85 MAE is a 0.22-year gain, and stating it that way makes the modest size obvious.
- 3. Beyond a single number.** Precision, recall, F1 and a full confusion matrix at threshold 0.5 are stored per model, because AUC alone hides which class is being missed.
- 4. Forward in time.** Random folds let a model learn from graduating classes that, in production, have not happened yet. The back-test trains on class $\leq$ cutoff and tests on later classes. On lacrosse, where this has been measured, HS→Pro fell from **0.869 to 0.706** — a 16-point drop. The basketball harness is written but its labelled scrape data is not committed, so on a fresh clone it reports the missing files and exits rather than writing fabricated numbers; the CV figures above are labelled as CV for that reason.
- 5. Calibration, not just ranking.** AUC says whether players are ordered correctly; it says nothing about whether "80%" means eighty percent. A 10-bucket out-of-fold **Expected Calibration Error (ECE)** is computed per model — predictions are sorted into 10 confidence buckets, each bucket's average prediction is compared to the rate that actually occurred, and the gaps are averaged by bucket size. Reported in percentage points. On lacrosse this caught a model at 24 pp ECE — predicting above 90% where the true rate was 5% — caused by class balancing; retraining without it brought ECE to **0.6 pp**.

HONEST
LIMITS

Every metrics file ships a `caveat` field that the product renders next to the number. **College** → **NBA**: the 0.910 AUC uses random NCAA players as negatives, so it mostly proves NBA players outproduced the average college player — not that it can separate good players who make it from good players who don't. **HS** → **NBA**: MaxPreps rebounding and assist data are sparse, coverage skews to 2006 onward, and 0.795 should be read as the honest answer to how much high-school box scores alone predict the NBA. **Longevity**: R^2 0.077 means career length is mostly noise given rookie-year features, so the product shows a range rather than a point estimate.

OmniSearch MULTI-AGENT REINFORCEMENT LEARNING

2026

Simulation platform for testing coordinated drone and ground-robot swarms in wildfire search-and-rescue. UC Berkeley MIDS capstone.

MODELS	HAPPO (heterogeneous-agent PPO, via HARL) as the primary policy; MAPPO and IPPO as controlled comparisons. Perception: YOLOv8s detectors plus a probabilistic detection model during RL.
TRAINING	VMAS wildfire scenario on real USGS/OpenStreetMap terrain, physical units, $\gamma = 0.99$, 900 steps @ 2 s = 30 min simulated per episode. 4 UAVs, 3 UGVs, 5–10 survivors on 1 km ² . BC warm-start and DAgger imitation available. Detectors fine-tuned on synthetic composites.
EVALUATION	100 held-out seeds per configuration against six scripted baselines (lawnmower, ant-colony, highest-confidence, random walk). Communication-dropout sweep 0–100%. Detectors evaluated against <i>real</i> datasets, not their own validation split.
RESULTS	Mission success 80% vs 72% (ant-colony) and 68% (lawnmower). Survivor confirmation ~10% faster. Under communication dropout the learned policy holds 76–82% through 70% loss while baselines fall ~14 points immediately. Detection: HERIDAL aerial 0.86 recall @ 0.50 precision; HIT-UAV thermal mAP50 0.923 real-trained vs 0.167 synthetic-only.

Predicts whether a multimodal prompt will break an LLM's safety guardrails *before* any content is generated, and which policy it would violate. UC Berkeley DATASCI 207.

MODELS	Baseline multinomial logistic regression ; random forest ; and a two-branch DNN (TensorFlow) — text+image TF-IDF → 256 → 128 ReLU with dropout → 128-d embedding, fused with a 16-d harm-score branch → 144-d → 64 → 17-way softmax.
TRAINING	48,000 prompts (JailBreakV-28K + 20K benign), 80/10/10 stratified. 26-d pre-generation feature vector: prompt length, roleplay and instruction-override markers, Detoxify harm score, OCR text, LLM image descriptions, FigStep flag. Dropout 0.3, early stopping patience 5, Adam, LR tuned 1e-4 vs 1e-5.
EVALUATION	Macro-average recall (false negatives are the risk), 5-fold stratified CV, per-category confusion matrices. Labels derived by a rule-based refusal classifier validated at 96% human agreement on 100 samples.
RESULTS	Recall 69.4% → 93.0% , accuracy 98.5%. Notably, adding image features to the linear model moved recall only 69.4 → 69.6, and random forest to 70.2 — multimodal features paid off only once the architecture could fuse them . Benign: 1.00 recall and precision. Malware and Financial Fraud > 0.97 F1. Long tail fails: Child Abuse 0.42 recall (12 training samples), Health misses 30% (10 samples).

DevDigest DEVELOPER TOOLS · APPLIED LLM

2026

Local-first AI pull-request reviewer whose engine cannot cite a line that is not in the diff. Also a course template: a minimal working starter plus eight lessons that each add one feature back.

MODELS	Provider-injected LLM — Anthropic, OpenAI, or OpenRouter . <i>No model trained</i> .
TRAINING	<i>Not applicable</i> . Quality comes from retrieval and constraint: a <code>repo-intel</code> index (ast-grep symbols, dependency-cruiser import graph, PageRank + git hotness) supplies a repo map as prompt context, computed once on clone.
EVALUATION	A mandatory grounding gate : every finding is checked mechanically against the diff and dropped if it cites no real changed line; the verdict score is recomputed from survivors rather than trusted from the model. 73 test files across five suites — hermetic units with a stubbed provider, testcontainers Postgres integration, deterministic browser e2e — each with its own CI workflow.
RESULTS	Working end to end: 36K lines of TypeScript across five packages. Untrusted diffs are fenced with an injection guard; structured output validated via Zod-derived JSON Schema with parse-and-repair.

Onpace MOBILE · APPLIED LLM

2026

Swim-coaching iOS app (App Store, v1.0.4). Every generative surface is bounded so the model proposes and the coach disposes.

MODELS	Claude Sonnet for plan generation, in a callable Cloud Function. Retrieval: Neo4j swim knowledge graph + vector search with text-embedding-3-small (1536-d). <i>No model trained</i> .
TRAINING	<i>Not applicable</i> . Hybrid graph + vector RAG. Anti-repetition: a rolling per-athlete history feeds the 4 most recent generated sessions back into the prompt (3 persisted forward) so plans progress instead of looping.
EVALUATION	<i>No formal offline evaluation</i> . Safety is enforced structurally — AI plans arrive as drafts into a coach's editor, adjustment can only modify an existing coach workout (preserving focus and authorship, dropping paddles on reported shoulder pain), and video review structures the coach's own narration rather than grading the swimmer.
RESULTS	Shipped to the App Store. Swimmer-reported outcomes from reviews: 100 m pace 1:45 → 1:38 in six weeks; one beginner from not completing a length to 400 m continuous.

Figures are drawn from committed metrics JSONs, capstone reports, and diagnostic outputs in each project's repository. Where a project reports both random-fold and temporal-holdout scores, the temporal number is the one quoted. Portfolio and full case studies available on request.